



Evaluating landscape connectivity in fragmented habitats: Cantabrian capercaillie (*Tetrao urogallus cantabricus*) in northern Spain



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ABSTRACT

Cantabrian capercaillie (*Tetrao urogallus cantabricus*) is listed as endangered according to IUCN criteria. The high degree of fragmentation and anthropogenic disturbances in capercaillie habitat suggests that habitat patterns may be related to decline of capercaillie populations in northern Spain. The objectives of this study are: (1) determining critical territories for the maintenance of capercaillie connectivity; and (2) evaluating the importance of public forests and their appropriate management to maintain the habitat connectivity for this species. This study is focused on northern Spain, where forest areas are critical for the maintenance of capercaillie. We applied connectivity methodologies based on morphological spatial pattern analysis (MSPA) and the probability of connectivity (PC). The results of the MSPA were incorporated into a standard GIS and compared with the spatial distribution of the public forest. Most of the valuable areas for connectivity were inside the public forests. Moreover, these public forests mainly form continuous features. Therefore, forest planning and management, mainly in public forest, should approach this problem including connectivity considerations and, more specifically, identifying the most critical forest sites for the maintenance of capercaillie habitat.

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1. Introduction

Capercaillie (*Tetrao urogallus*) populations have declined throughout most of the species range, especially in central and southern Europe (Storch, 2000). This trend has been observed in the Black Forest in southwestern Germany, which holds the largest capercaillie population in central Europe outside the Alps (Braunisch and Suchant, 2007) or in other colonies such as the southern remnant population in the Rhodope Mountains in Bulgaria (Graf, 2006). At a national level, 14 of the 18 grouse species are red-listed in at least one country (Storch, 2007b). Populations at the southern edge of a species' range and in densely populated regions are most often red-listed; habitat degradation, loss and fragmentation due to human land use activities are the major threats to viability of capercaillie populations, availability of habitat due to climate change, land use fragmentation (Klaus, 1994; Storch, 2007a). On local and regional scales, many populations of capercaillie are declining and threatened with extinction (Storch, 2007b).

The Cantabrian subspecies of capercaillie (*Tetrao urogallus cantabricus*) is listed as endangered according to IUCN criteria (Storch et al., 2006). At the beginning of the 2000s it was estimated that there were about 600 Cantabrian capercaillies present in an area of 1700 km² in the Cantabrian Mountains of northern Spain (Martí and Moral, 2003; Storch, 2007b). The population had a female: male sex ratio of 1:1.6 (Morán-Luis et al., 2014). Compared to a historic range of 3500 km², the area of occupancy has declined by >50% in the last 20–30 years (Martí and Moral, 2003; Quevedo et al., 2006b).

Capercaillie requires large tracts of mature forest, and are thus sensitive to landscape-level habitat alteration (Storch, 1995). Therefore, integrating forestry practices and capercaillie habitat preservation is a major conservation challenge (Storch, 2007b). The high degree of fragmentation and anthropogenic disturbances in the habitat of Cantabrian capercaillie suggests that changes in habitat patterns may be related to the population decline (Gómez-Manzanedo et al., 2009; Quevedo et al., 2006b). Moreover, improving connectivity between patches and populations is one of the main measures proposed for conservation of the subspecies (Espinosa et al., 2004; Storch, 2007b). This implies that forest planning and management should approach this problem by explicitly including connectivity considerations and, more specifically, by

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identifying the most critical forest sites for the maintenance of capercaillie habitat connectivity (Pascual-Hortal and Saura, 2008).

In the Cantabrian Mountains, most of the capercaillie habitats are located in public forests. This legal status guarantees protection and maintenance of the forests and their fauna (Espinosa et al., 2004; Garcia et al., 2005) that may lead to an improved conservation status of this species by connecting its populations, facilitating dispersal of individuals and minimizing mortality risk. In this context, it is essential to identify which areas are critical habitats for connectivity since loss of forest habitat would probably lead to exponential increases in isolation of the remaining fragments (Garcia et al., 2005). This situation could be particularly important for the isolated populations of endangered forest vertebrates still present in the Cantabrian Mountains but highly sensitive to habitat degradation, such as capercaillie (Obeso and Bañuelos, 2003).

In this context, a wide number of connectivity indices have been developed (Bunn et al., 2000; Calabrese and Fagan, 2004; Jordan et al., 2003). However, several of these indices do not allow the inclusion of any qualitative difference between adjacent landscape patches in the computation of the index (Acosta et al., 2003), or they present serious limitations that make them inadequate for effectively integrating connectivity in forest conservation planning (Pascual-Hortal and Saura, 2006).

Recent studies (Estreguil and Mouton, 2009; Saura et al., 2011) propose new methodologies based on the use of: (1) Morphological Spatial Pattern Analysis (MSPA) of forest habitats (Soille and Vogt, 2009; Vogt et al., 2007a,b); and (2) the probability of connectivity (PC) (Pascual-Hortal and Saura, 2006; Saura and Pascual-Hortal, 2007; Saura and Rubio, 2010). These aspects offer considerable synergies and potential relevant benefits for forest planning at a variety of scales, and address some of the concerns and limitations that forest managers face when identifying key structural connecting elements (Saura et al., 2011).

The objectives of this study are: (1) determining critical areas for the maintenance of connectivity of Cantabrian capercaillie habitats; and (2) evaluating the importance of public forests and their appropriate management to maintain the habitat connectivity for this species.

2. Material and methods

2.1. Study area

For this study, we adopted a potential territory for the Cantabrian capercaillie based on land use and species distribution's map criteria. The public forests of the Asturias and León regions (Spain), located in the Cantabrian Mountains (Fig. 1), are of particular interest because it is expected that management plans could be applied relatively easily.

2.2. Method

The general outline of the method is shown in Fig. 2. The method includes derivation of a hybrid land-cover map by combining standardized land-cover classes from CORINE Land Cover 2006 (Pérez-Hoyos et al., 2012) with higher spatial resolution data from SIOSE (Sistema de Información sobre Ocupación del Suelo de España; Spanish land use and land cover information system) (Valcárcel et al., 2008).

2.2.1. Phase 1: potential territory assessment

The first step of the method is to determine the area of potentially suitable habitat. The vast majority of capercaillie populations inhabit boreal coniferous forests (Storch, 2007b). However, Cantabrian capercaillie shows substantial ecological differentiation associated with its distinctive habitat (Blanco-Fontao et al.,

2009). The Cantabrian population lives in mature beech (*Fagus sylvatica*) and mixed beech/oak (*Quercus* spp.) forests with bilberry (*Vaccinium myrtillus*) (Quevedo et al., 2006a; Storch, 2007b), and also in meadow habitats, heathlands and shrublands (Espinosa et al., 2004).

For this study areas that fulfill the following two conditions were defined as suitable habitat: (i) areas that were less than 5 km from the actual distribution area of capercaillie in the Cantabrian Mountains (MARM, 2007); and (ii) areas that were labelled with the land cover classes CLC3.1 "Forest" (including broad-leaved forest, coniferous forest and mixed forest), and CLC3.2 "Shrub and/or herbaceous vegetation associations" (including natural grassland, moors and heathland, sclerophyllous vegetation and transitional woodland/shrub). This potential suitable habitat has a total area of 8274.2 km² (Fig. 3). Of this area 7256.6 km² is public forest.

2.2.2. Phase 2: morphological spatial pattern analysis (MSPA)

The second step of the method is the connectivity analysis through the Morphological Spatial Pattern Analysis (MSPA) (Soille and Vogt, 2009); MSPA has been implemented into a stand-alone freeware called GuidosToolbox available at <http://forest.jrc.ec.europa.eu/>. MSPA performs a segmentation on a binary map at pixel level, resulting in seven mutually exclusive spatial pattern classes: Core, Islet, Perforation, Edge, Loop, Bridge, and Branch (Soille and Vogt, 2009).

Amongst other parameters, the MSPA segmentation process may be fine-tuned by a user-defined edge width parameter. In this study three edge widths (EW) were assessed for a number of important capercaillie habitat areas, ranging in size from 132 ha to 1207 ha (Gjerde and Wegge, 1987; Schroth, 1991; Storch, 1995). The identified habitats had a mean area of 500 ha (Obeso, 2004). For this estimation, we assumed EW was equivalent to the radius of an ideal circular vital area (Table 1).

2.2.3. Phase 3: connectivity index

The method to measure connectivity in this study uses the Probability of Connectivity (PC) index calculated with an adapted version of the software Conefor Sensinode (Saura and Torne, 2009) <http://www.conefor.org/>.

PC is recommended as a connectivity measure because it reacts meaningfully to habitat loss and network fragmentation (Saura and Pascual-Hortal, 2007). PC also possesses the richest connection model of the measures in widespread use in connectivity evaluation (Laita et al., 2011).

The method is based on topology (inter-patch distances) and patch attributes (area) for forest dwelling species with a specific dispersal ability. The index combines landscape graph theory, a probabilistic connection model and the habitat availability concept. As a result, a landscape graph is made up of a set of nodes and links between nodes (Estreguil and Mouton, 2009). Each link between two patches is characterized by a probability of dispersal, obtained as a function of distance.

For our study, we need to define a typical dispersal distance for Cantabrian capercaillie. The data reported in the literature present averages ranging between 2.7 km in Sweden (Kempff, 1982) and 1.5 km in Norway (Rolstad et al., 1988). More recent studies show mean dispersal distances of 2.4 km in Norway and 2.2 km in Russia (Hjeljord et al., 2000) or median values of 11 km in Scotland (Moss and Picozzi, 1994). However, there is little available data on dispersal distance for Cantabrian capercaillie (Abajo, 2007). We assumed an average ranging distance of 2.3 km, following Gómez-Manzanedo's (2009) study of capercaillie in the Cantabrian Mountains.

From the seven pattern classes provided by MSPA only the classes Core and Bridge are relevant for a graph-theory investiga-

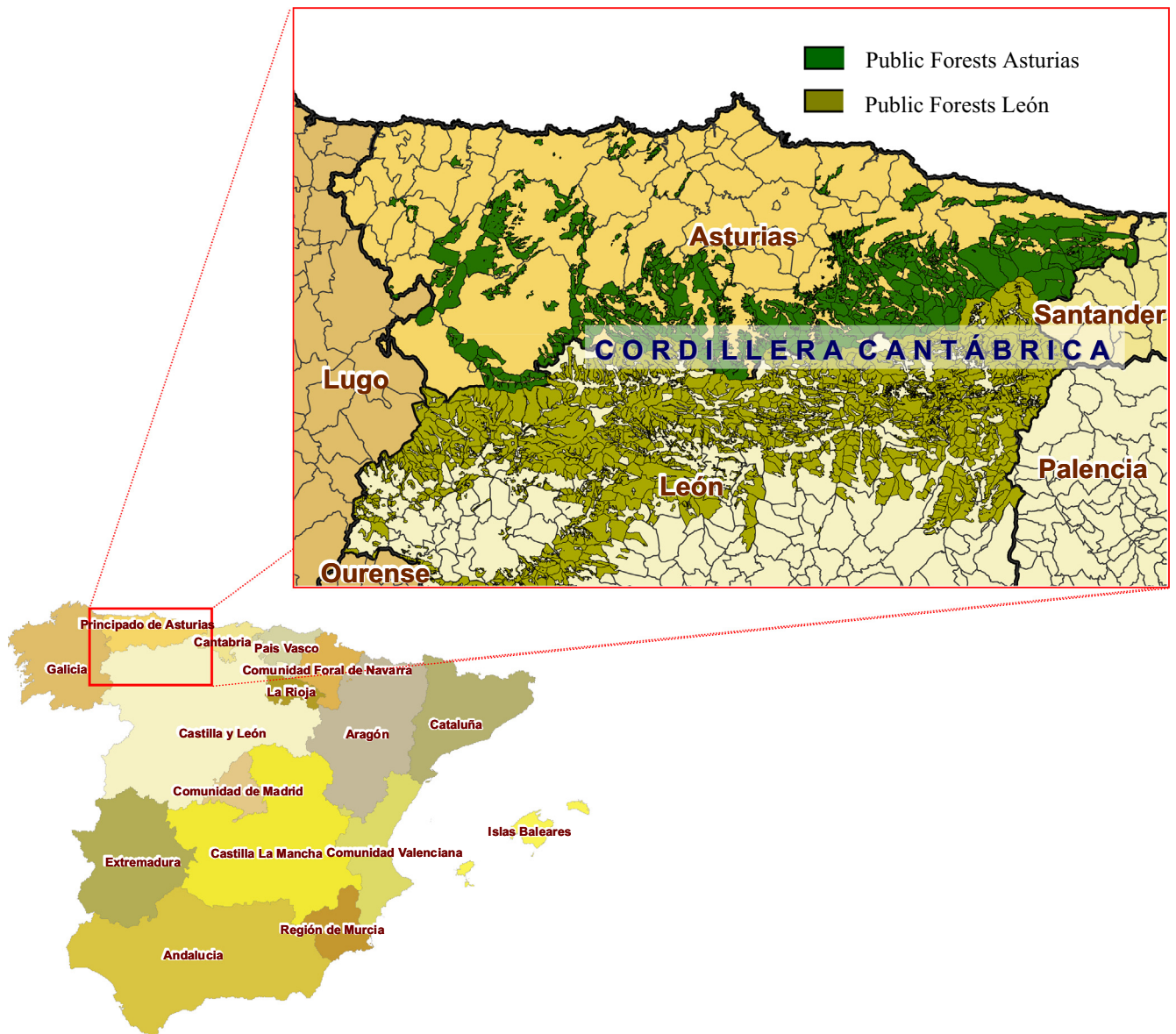


Fig. 1. Public forest of Asturias and León regions.

tion, where a Core corresponds to a Node and a Bridge corresponds to a Link (Saura et al., 2011). Therefore, the PC values for the MSPA detected Core and Bridge classes allow for the evaluation of the importance of each individual landscape element for maintaining connectivity in the potentially suitable habitat area.

PC values for Cores are calculated for the assumed dispersal distance, meaning they are connectivity providers of focal habitat area depending on that distance. On the other hand, PC importance for Bridges is calculated for an unlimited dispersal distance since corridors are expected to effectively facilitate dispersion through these corridors.

Finally, PC results are incorporated into a standard Geographical Information Systems (GIS) that provides their spatial distribution on the public forest of Cantabrian Mountains.

3. Results and discussion

3.1. MSPA

In the study area and for an EW of 25 pixels, 25.3% of the potential habitat was classified as Core habitat (a total area of

209,519 ha, with 747 Core patches) and 35.4% as Bridges (Table 2 and Fig. 4A). For an EW of 50 pixels, 6.38% of the potential habitat was classified as Core habitat (52,778 ha, with 197 Cores) and more than 70% as Bridges (Table 2 and Fig. 4B). For an EW of 100 pixels (Fig. 4C) there were nine Core areas (5000 ha) located only in the western part of the study area. This is not in accordance with the known current distribution of capercaillie in the Cantabrian Mountains, and therefore the pattern of Core areas with an EW of 100 pixels was not further analyzed.

3.2. PC for Cores

PC values of Cores for an EW of 25 pixels are presented in Fig. 5. The area and PC values for the six most important Cores are shown in Table 3. The biggest Core (C1) was in Asturias region, in the western part of the study area and due to its large area (over 40,000 ha) it can highly contribute to the conservation of capercaillie and facilitate connectivity among western Capercaillie populations and northern populations of the Galicia region (not included in the study area). This result is in accordance with other studies highlighting that this subspecies requires extensive forest

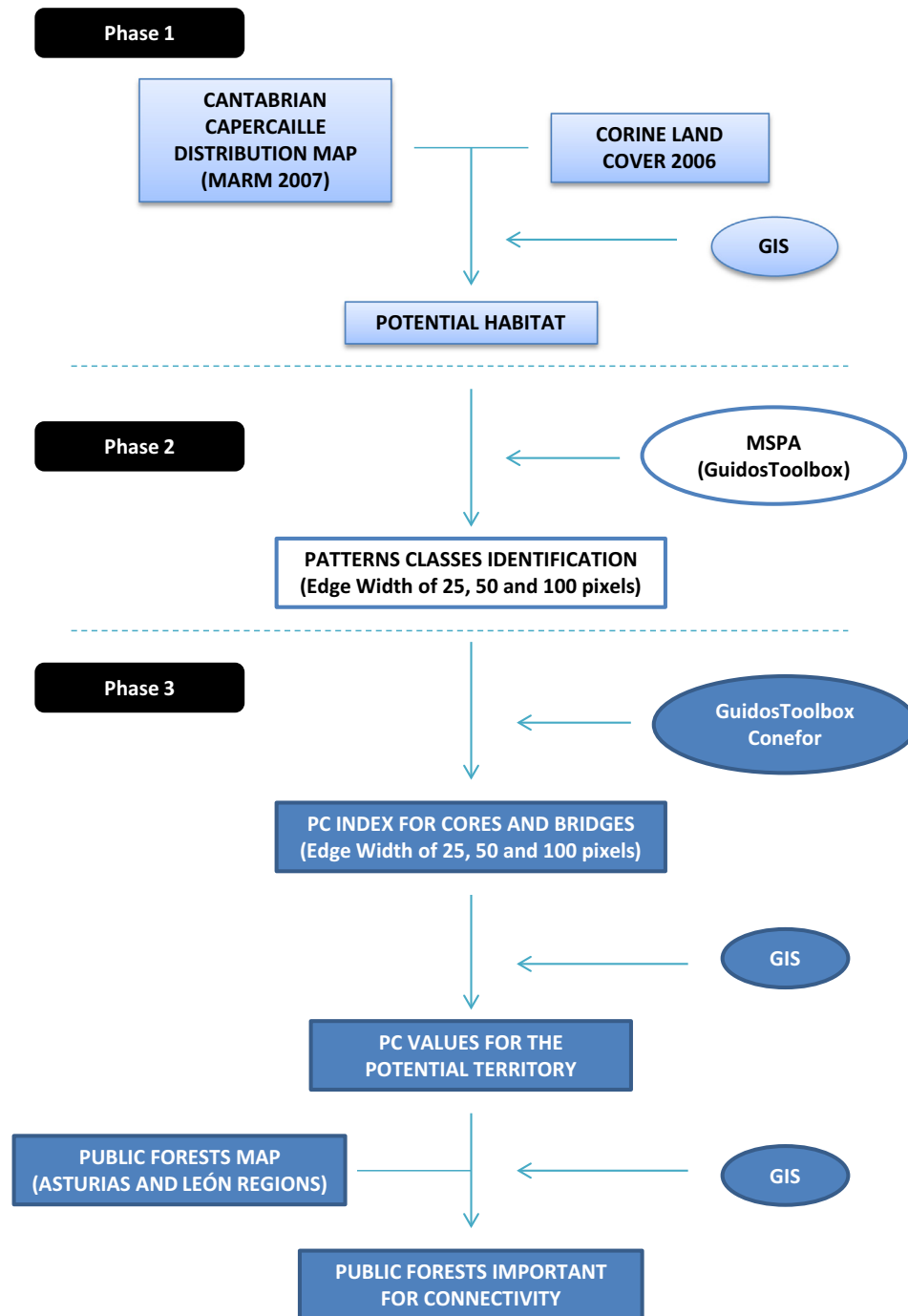


Fig. 2. General outline of the method.

areas with a high proportion of tree cover (Quevedo and Bañuelos, 2008; Quevedo et al., 2006b). This area would include not only leks but also refuges, broadly understood as including suitable areas in terms of feeding, competition, shelter from predators, among several other relevant aspects for the survival of a population (Berryman and Hawkins, 2006).

Although relatively small, Cores C2 and C3 are important for connectivity in the center of the area ($PC_{25} = 12.63$ and $PC_{25} = 8.75$) and can improve the current state of capercaillie habitat in the Asturias region. Recent Asturian data show that capercaillie populations in the central part of the Cantabrian Mountains are almost gone, with just a few isolated spots where

capercaillie may still be seen (Bañuelos and Quevedo, 2008). C4, C5 and C6 are relatively large and can also be considered as suitable areas due to the high proportion of forests (Bañuelos et al., 2008). Our analysis also revealed a lack of Core patches in the central León region due to the deforestation caused by the traditional use of the land for livestock (García et al., 2005; Gil, 2007).

Table 4 shows the area and PC values of the six most important Cores for an EW of 50 pixels. These Cores are smaller than Cores for an EW of 25 pixels (C1 to C6), except for C7. PC values are also lower than in the previous analysis (EW25). In addition, this analysis did not identify important Cores in the central part of the Asturias and León regions. This may indicate that the dispersal dis-

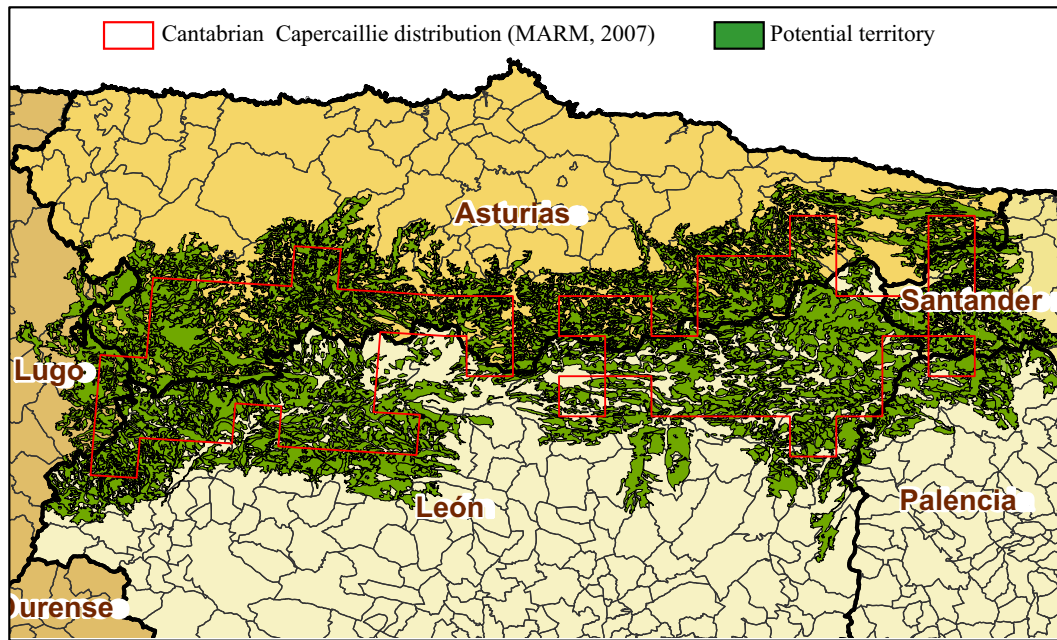


Fig. 3. Potential territory.

Table 1

Vital area and edge width (EW).

Vital area (ha)	Radius (m)	Number of Pixels (Raster 25 m)	EW (pixels)
132	648	26	25
500	1262	50	50
1207	1960	78	100

Table 2

MSPA results: Cores and Bridges.

Edge width (pixel)	Number of Cores	Total Core area (ha)	Mean area per Core (ha)	Total Bridge area (ha)
25	747	209519.56	280.48	292795.19
50	197	52787.00	267.91	579784.12

tance used to measure the importance of Cores for an EW of 50 pixels needs more accurate assessment.

We can compare these PC values of Cores with figures from other studies, which show results of 9.19% (maximum) and 45.59% (sum) both for the most optimistic scenario (PC-infinite) in the Spanish province of Segovia (Castilla and León region); and 11.7% (maximum) and 37.44% (sum) for the Boyal forest land (also in the Castilla and León Region, Spain) for the same scenario (Saura et al., 2011).

3.3. PC for Bridges

The highest PC value (55.23) for Bridges was calculated for the area B1 (Table 5). The area of B1 is over 25,000 ha and connects with the center of the study area (Fig. 5). Another important Bridge is B2 (PC = 12.63). Both B1 and B2 connect the eastern and western capercaillie's habitats with the central Cores C2 and C3.

The six important Bridges consist of 43.12% CORINE Class 3.1 (Forest) and 57.88% CORINE Class 3.2 (Shrub and/or herbaceous vegetation associations). Therefore, there is a large area (97,226.6 ha) of natural grassland, moors and heathland, sclerophyllous vegetation and transitional woodland/shrub with

importance for the connectivity. This result supports other studies related with Cantabrian capercaillie that suggest that the non-forest matrix should also be included in management plans, as a direct source of food and shelter, and an important feature that may increase the connectivity between forest patches (Quevedo et al., 2006a). Moreover, this unique capercaillie population has been studied and managed largely from a lek-centered perspective that has somewhat neglected the importance of habitats other than old forest, such as canopy gaps or tree line habitats with rich heath cover, which could be important in other stages of the capercaillie's life cycle (Bañuelos et al., 2008; Quevedo et al., 2006a).

For an EW of 50 pixels, the analysis assigned the same importance to all Bridges. This does not match with the known distribution of capercaillie.

Our results can be compared with those of Saura et al. (2011), which used PC-infinite index, which expresses forest habitat network in which movement through Bridges is considered as completely feasible with the maximum probability independent of the distance that needs to be traversed through those bridges to get from one Core to another. This index of the most optimistic scenario (PC-infinite) for Bridge importance in the Spanish province of Segovia was 16.67% (maximum) and 42.36% (sum); and 25.87% (maximum) and 50.94% (sum) for the Boyal forest land.

3.4. PC for public forest

The results of the PC analysis (EW 25 pixels) were compared with the location of publicly owned forest. Fig. 6 shows the public forest of the Asturias and León regions important for Cores, while Fig. 7 shows the public forest of Asturias and León regions important for Bridges.

Most of the valuable areas for connectivity are inside the public forests (Table 6). Moreover, these public forests mainly form continuous features (Figs. 6 and 7). These results allow actions to be taken at a landscape scale.

Nevertheless, the mean size of public forest area is 0.085 km² in the Asturias region and 0.1 km² in the León region. Taking into

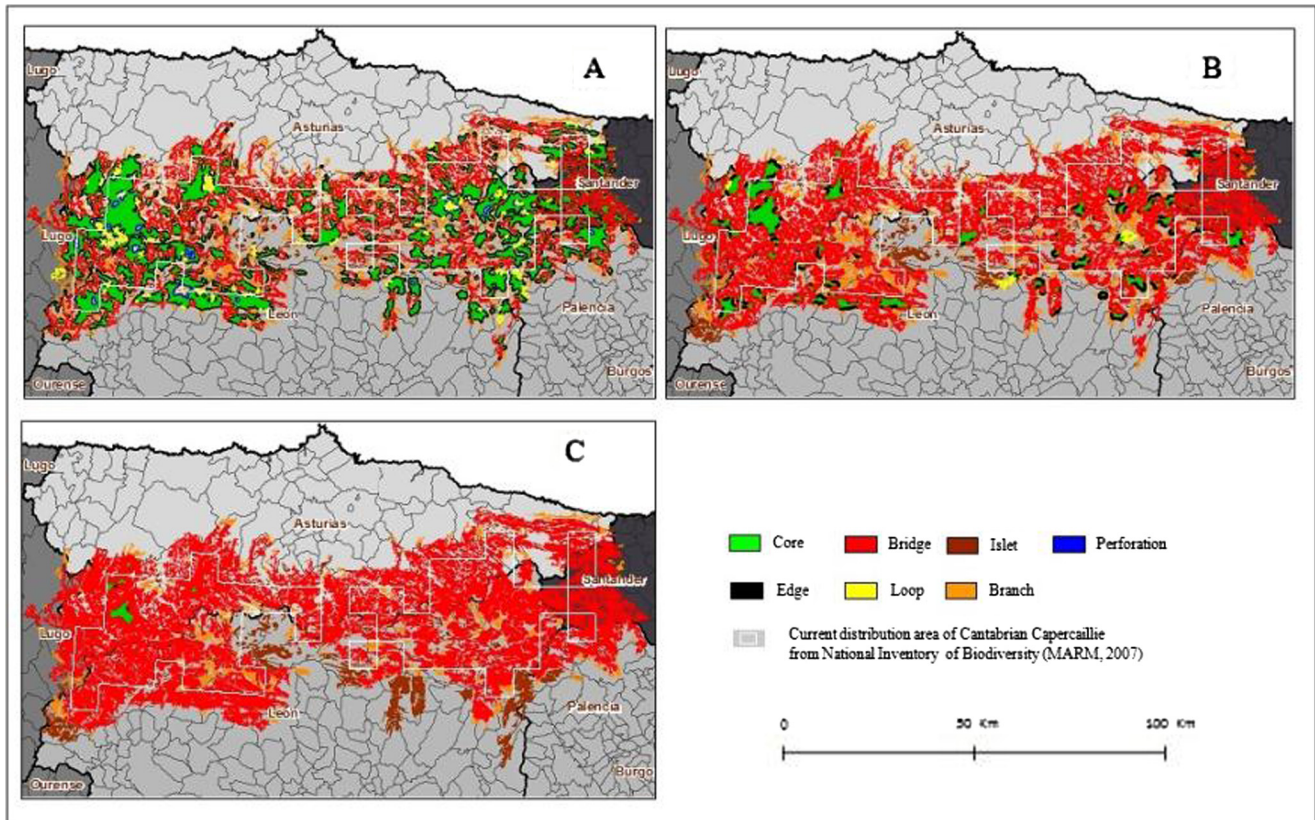


Fig. 4. MSPA classes resulting for an edge width of (A) 25 pixels, (B) 50 pixels and (C) 100 pixels.

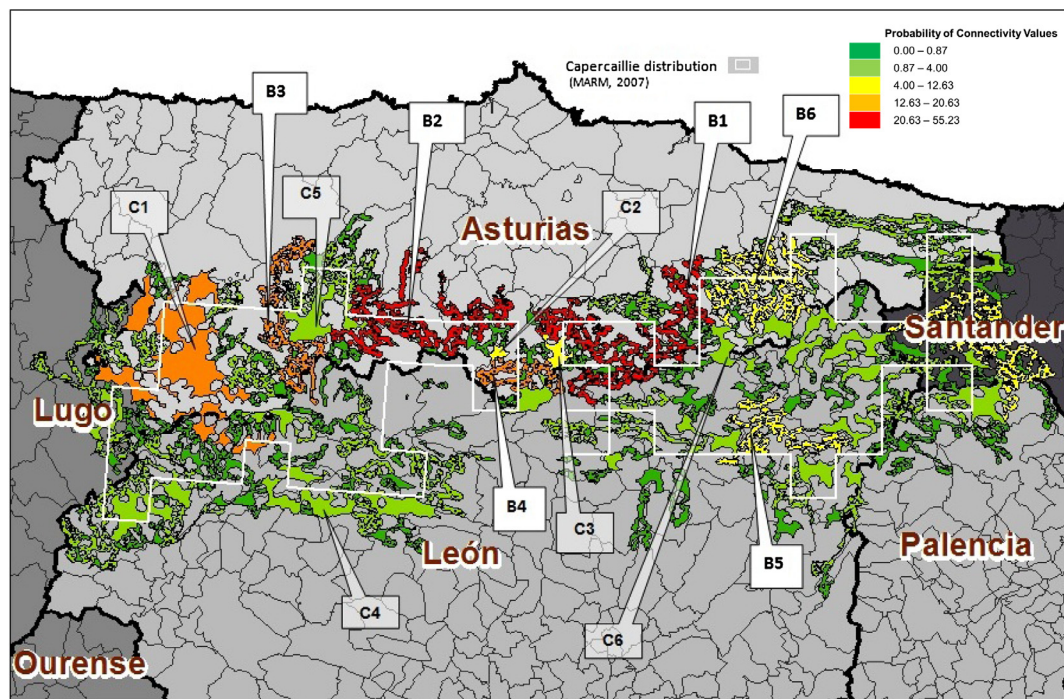


Fig. 5. PC values for Cores classified with EW of 25 pixels and a dispersal distance of 2.3 km.

account that the high proportion of forest cover in a neighborhood of 50 km² is one of the more significant variables for the assessment of suitable Cantabrian capercaillie habitat (Quevedo and Bañuelos, 2008), the need of management at the landscape scale is reinforced.

Public forests located in the center of the Asturias Region are of great importance for capercaillie connectivity due to the presence of the Cores C2 and C3, and the Bridges B1, B2 and B4. This result is in accordance with other studies (Quevedo and Bañuelos, 2008) that looked at the habitat quality for Cantabrian capercaillie.

Table 3

Important Core areas: identification, area and PC values for an EW of 25 pixels.

ID	Area (ha)	PC
C1	41,040.87	17.35
C2	878.43	12.63
C3	2350.30	8.75
C4	9417.93	3.99
C5	8550.12	3.43
C6	7327.87	2.94

Table 4

Important Core areas: identification, area and PC values for an EW of 50 pixels.

ID	Area (ha)	PC
C7	8327.41	2.62
C8	5691.36	1.79
C9	2738.15	0.86
C10	2354.21	0.74
C11	2159.04	0.68
C12	1474.55	0.65

Table 5

Important Bridge areas: identification, area and PC values for an EW of 25 pixels.

ID	Area (ha)	PC
B1	25,501.00	55.23
B2	22,626.18	54.25
B3	13,060.50	20.62
B4	6347.62	19.01
B5	10,369.12	7.96
B6	19,322.18	6.98
Total area	97,226.60	

The areas C1, C5 and B3 also include a large proportion of private land, and actions should at least be aimed at preserving and improving these public forests. Conservation measures should focus on the remaining large forest fragments in the area and the understory; particularly bilberry (*Vaccinium myrtillus*) should be favored in these areas, preserving the old-forest structure (Quevedo et al., 2006a).

The central part of the León's study area lacks forested areas wide enough to provide Cores and Bridges that allow habitat and connectivity for Cantabrian capercaillie. This area is deforested due to the traditional use of land for livestock (García et al., 2005; Gil, 2007). Therefore, a variety of measure are suggested in order to improve connectivity in this area. The main measures are: (1) creation of clearings and appropriate forest structure that provide suitable capercaillie habitat (Moss and Picozzi, 1994; Sachot, 2002); (2) strict protection of areas where capercaillie are known; (3) avoidance of creation of new forest roads and new areas of forest exploitation; (4) control of deer and wild boar populations at local and landscape scales; (5) restrict keeping of livestock in forest and adjacent areas to avoid conflict with capercaillie (Velázquez et al., 2008).

Moreover, because of the untypical capercaillie habitat in this region, it will be of great interest to implement locally adapted conservation measures (Blanco-Fontao et al., 2009; González et al., 2009). These could include measures to specifically protect bilberry from overgrazing by livestock, due to the strong reliance on bilberry by Cantabrian capercaillie (Blanco-Fontao et al., 2009), and measures to improve bilberry productivity (Tolvanen, 1994).

4. Conclusions

The Morphological Spatial Pattern Analysis (MSPA) and the PC index used for this study enabled the identification of important

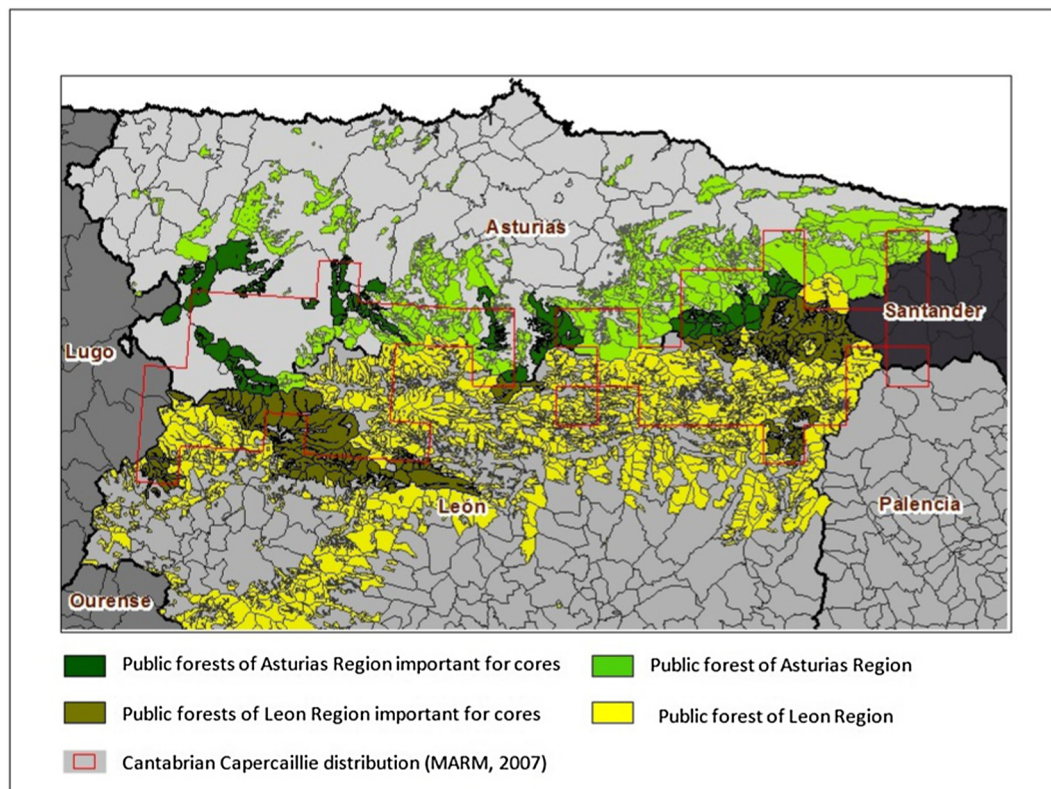


Fig. 6. Public forest of Asturias and León regions important for Cores.

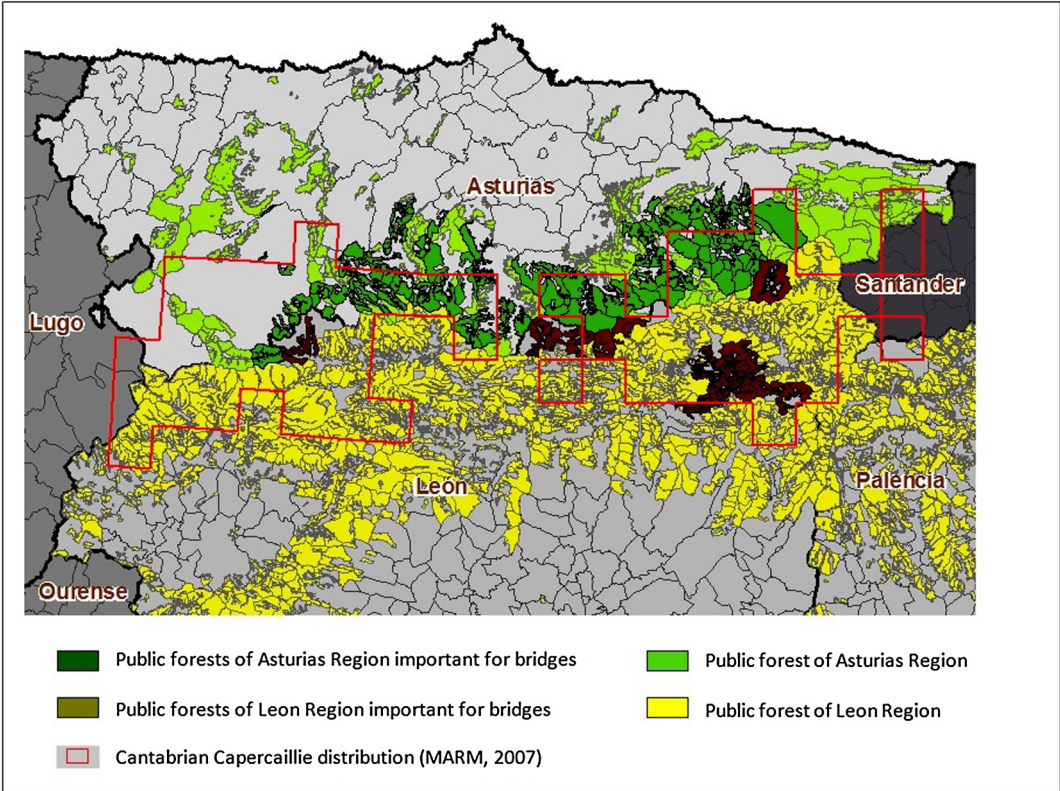


Fig. 7. Public forest of Asturias and León regions important for Bridges.

Table 6
Importance of Cores and Bridges by percentage of inclusion in public forests.

% Core or Bridge area/public forest area	Cores and Bridges of importance for connectivity
0–25	–
25–50	C1, C5, B3
50–75	C2, C3, C9, B2
>75	C4, C6, C8, C9, C10, C11, C12, B1, B2, B4, B5, B6

habitat areas for Cantabrian capercaillie and how well they are connected. Much of the area is public forest where conservation actions should be implemented.

The Asturias region contains Core patches that are important for the conservation of capercaillie. Especially those located in the center can improve the current state of capercaillie habitat in the Asturias region.

The land use of the more important Bridge features suggests that the non-forest matrix should also be included in management plans, as a direct source of food and shelter, and an important feature that may increase the connectivity between forest patches.

Most of the valuable areas for connectivity are inside the public forests of the Asturias and León regions. These forests mainly form continuous features that allow a landscape management approach. This is a relevant result since Cantabrian capercaillie is highly sensitive to landscape-level habitat alteration.

Finally, more information is needed about dispersal distance and vital area of this subspecies, given that these are key parameters of the methodology and incongruent outcomes with the current distribution of the subspecies emerged from our analysis for EW of 50 and 100 pixels.

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